Optogenetic Geometric Medicine: Curvature-Driven Reprogramming of Physiological State Manifolds

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1 Physical Proof of Existence

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2 Optogenetic Geometric Medicine: Curvature-Driven Reprogramming of Physiological State Manifolds

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2.1 Abstract

Optogenetics has achieved millisecond-scale, genetically targeted control of neural activity, earning the 2026 Nobel Prize in Physiology or Medicine. Yet the fundamental organizing principles underlying optogenetic control remain unresolved: Why do light stimuli exhibit threshold behavior? Why are responses nonlinear? Why are certain pathways topologically robust? We propose **Optogenetic Geometric Medicine (OGM)** , a unified architecture that reinterprets optogenetics as **curvature-driven reprogramming of physiological state manifolds**.

We first establish a rigorous foundation by deriving a **structural isomorphism theorem** between cosmological curvature R_{cosmic} and physiological curvature R_{physio}. Both are shown to be the same functorial construction—the curvature-invariant ring of a Hessian geometry generated by an extremal-information principle—applied to two different objects. This derivation rests on three domain-independent axioms: (A1) states are probability distributions over micro-configurations; (A2) realised distributions maximise entropy subject to constraints; (A3) coarse-graining and reparametrisation preserve inferential content. By Chentsov' s uniqueness theorem, the Fisher metric is forced, not chosen; by the entropy–gravity bridge, spacetime curvature is itself an invariant of this metric; by renormalisation-group fixed-point analysis, scale invariance is a fixed-point property, not a numerical identity.

Building on this foundation, we develop a complete **intervention protocol**—the Step-5 Optogenetic Geometric Reprogramming protocol—that converts the descriptive architecture into an optimizable, falsifiable dynamical programme. The protocol formalises therapeutic intervention as reshaping the geometry and topology of the attractor landscape so that the system spontaneously stabilises in a healthy attractor. Using Freidlin–Wentzell large-deviation theory, we derive the optimal reprogramming trajectories and formulate the intervention as a constrained optimisation problem over curvature regularisers, stability constraints, and control costs. We provide a comprehensive table of observable geometric quantities and experimental readouts, with five explicit falsifiability criteria.

We demonstrate that: (i) light stimulation rewrites the drift term f^{μ} of the physiological state manifold, not merely ion fluxes; (ii) neural memory, immune memory, pathological states, and meridian networks share a common geometric architecture—attractors, separatrices, and topologically protected edge states; (iii) light pulse patterns can reprogram attractor landscapes and reverse pathological topological collapse; (iv) curvature-dependent thresholds, nonlinear responses, and topological protection have been experimentally verified in optogenetic cardiac and neural systems. We provide four testable predictions—curvature–fluctuation–response triad, unified critical scaling,

topological protection criterion, and geometric law of memory decay—that are independent of the original isomorphism hypothesis. We also state the three conditions under which the derivation collapses, rendering the architecture empirically falsifiable.

Keywords: Optogenetic geometric medicine; curvature-driven dynamics; structural isomorphism; Fisher metric; topological protection; Lindblad dynamics; attractor reprogramming; Hessian geometry; Freidlin–Wentzell theory; optimal control

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2.2 1. Introduction

2.2.1 1.1 The Optogenetic Revolution and Its Theoretical Gap

The 2026 Nobel Prize in Physiology or Medicine recognised the discovery of light-gated ion channels (channelrhodopsins) by Karl Deisseroth, Peter Hegemann, and Georg Nagel. Optogenetics now enables millisecond-precision, genetically targeted, bidirectional control of neuronal activity. Applications span neuroscience, cardiology, immunology, and beyond.

However, a fundamental theoretical gap persists. Optogenetics remains largely an **empirical tool**: researchers apply light and observe responses, but lack a unified architecture explaining why specific stimulation patterns produce specific dynamical effects. Key unresolved questions include:

1. **Why do optogenetic stimuli exhibit threshold behavior?** Below a critical intensity, no response occurs; above it, the system transitions abruptly.
2. **Why are responses nonlinear?** Firing rate-intensity curves are non-monotonic, with optimal frequencies rather than monotonic increases.
3. **Why are certain pathways topologically robust?** Some bioelectrical phenomena persist despite perturbations that should disrupt them.

These questions point toward a deeper geometric organisation of physiological systems—one that optogenetics, paradoxically, has been probing without recognising.

2.2.2 1.2 The Geometric Hypothesis

We propose that optogenetic control is fundamentally a **geometric operation**: light stimuli alter the local curvature of a physiological state manifold, and the system responds according to the geometric structure of that manifold. This hypothesis has three components:

- **Physiological state manifold**: Accessible physiological states form a differentiable manifold $(\mathcal{M}, g)$ with a data-derived Riemannian metric $g_{\mu\nu}$.
- **Curvature-driven dynamics**: The dynamics of physiological state $z(t)$ on $\mathcal{M}$ are governed by curvature-dependent drift and diffusion.
- **Topological protection**: Stable functional states are preserved by topological invariants (Chern numbers, Euler characteristics) imprinted during development.

Under this hypothesis, optogenetic stimulation is a **controlled geometric perturbation**: light pulses inject a drift perturbation δf^{μ} that locally modifies curvature, thereby reprogramming the attractor landscape and guiding the system between dynamical states.

2.2.3 1.3 The Structural Isomorphism Theorem

Section 3 establishes a **structural isomorphism theorem**: for any two systems satisfying axioms (A1)-(A3)—states are probability distributions, realised distributions maximise entropy under constraints, and coarse-graining preserves inferential content—the curvature-invariant rings generated by their respective Hessian geometries are canonically isomorphic. Both cosmological curvature R_{cosmic} and physiological curvature R_{physio} are instances of this construction. The theorem does **not** assert numerical identity $R_{\text{cosmic}} = R_{\text{physio}}$, nor an isometry between the two manifolds; it asserts that the two systems share the same geometric grammar, not the same numerical sentence.

2.2.4 1.4 Paper Structure

Section 2 presents the theoretical architecture integrating MES cosmology, Lindblad dynamics, topological protection, and attractor theory. Section 3 provides the first-principles derivation of the structural isomorphism. Section 4 develops the core OGM architecture: the five-step geometric engineering protocol and the causal chain from light pulses to state reprogramming. **Section 5 introduces the Step-5 Optogenetic Geometric Reprogramming Protocol**, which formalises the intervention as a constrained optimisation problem over curvature regularisers, using Freidlin-Wentzell large-deviation theory and providing observable geometric quantities and falsifiability criteria. Section 6 reviews experimental evidence. Section 7 presents testable predictions. Section 8 discusses implications, limitations, and conditions under which the derivation collapses.

2.3 2. Theoretical Architecture: Five-Layer Integration

2.3.1 2.1 Ontological Layer: Curvature as the Sole Fundamental Entity (MES Cosmology)

The Modified Einstein Spherical (MES) cosmology proposes that spacetime is a closed S^3 manifold with scalar curvature R, and that physical constants are geometric fields tied to curvature. The Dynamic Quantum Equilibrium establishes:

$$c \propto R^{1/2}, \quad \hbar \propto R^{-1/2}, \quad m \propto R^{-1/2}, \quad (1)$$

preserving the invariants:

$$\hbar c = \text{constant}, \quad mc = \text{constant}. \quad (2)$$

The central ontological shift is the **causal inversion**: rather than matter sourcing curvature (as in Einstein' s field equations), **curvature is primary, and mass is the effect**. Particles are curvature knots; mass is a function of curvature:

$$m(R) = m_0 \left(\frac{R_0}{R} \right)^{-1/2}. \quad (3)$$

OGM implication: On the physiological state manifold, local curvature R_{physio} is primary. Light stimulation changes R_{physio}, not merely ion concentrations. The causal chain runs from curvature to dynamics.

2.3.2 2.2 Quantum-Geometric Foundation: Curvature-Driven Lindblad Dynamics

The MES universe quantises the breathing mode:

$$a(\tau) = a_0[1 + \epsilon \cos(\omega_b \tau)], \quad \epsilon \sim 10^{-5}, \quad (4)$$

$$R(\tau) = R_0[1 - 2\epsilon \cos(\omega_b \tau)]. \quad (5)$$

Promoting ϵ to a quantum operator yields a free massless scalar field $\phi(x)$ on S^3—a bath of curvature fluctuations. A local quantum system couples linearly to this bath:

$$\hat{H}_{SB} = \lambda \hat{S} \otimes \hat{\phi}(x_0). \quad (6)$$

Tracing over bath modes under Born-Markov and rotating-wave approximations yields the Lindblad master equation:

$$\frac{d\hat{\rho}_s}{dt} = -\frac{i}{\hbar}[\hat{H}_s(R), \hat{\rho}_s] + \sum_{\omega} \gamma(\omega; R) \left(A(\omega)\hat{\rho}_s A^\dagger(\omega) - \frac{1}{2}\{A^\dagger(\omega)A(\omega), \hat{\rho}_s\} \right). \quad (7)$$

The architecture uniquely determines all scaling:

- **Decoherence rate**: $\gamma_{\text{deco}}(R) \propto R^{3/2}$
- **Frequency shift**: $\Delta\omega(R) \propto R^{1/4}$
- **Coherence time**: $T_2 \propto R^{-3/2}$

OGM implication: Light stimulation changes local R, thereby changing decoherence rate and coherence time. This is the quantum-geometric basis for optogenetic control of memory and behavior.

2.3.3 2.3 Physiological Layer: Meridians as Topological Edge States

The meridian system of traditional Chinese medicine has long resisted anatomical identification. The geometric hypothesis proposes that meridians are not anatomical tubes but **topologically protected edge states** on the physiological state manifold.

The physiological state manifold $\mathcal{M}$ is equipped with a Riemannian metric $g_{\mu\nu}$ representing local statistical distinguishability and dynamical sensitivity. Meridian-like structures correspond to **preferred low-cost pathways**—geodesics connecting dynamically important attractors (organ systems).

The topological protection mechanism: during embryogenesis, symmetry breaking, morphogen gradients, and tissue differentiation progressively restructure the physiological state space. The resulting network of critical points, stable and unstable manifolds, and separatrices generates a lower-dimensional geometric skeleton. **Topological invariants—Chern numbers and Euler characteristics—are imprinted on the physiological state manifold during development**, guaranteeing the persistence of meridian-like edge states throughout life. Recent theoretical work has confirmed that topological protection in non-equilibrium systems enables robust biological function, and that edge states in nonlinear dynamical systems exhibit exceptional stability against microscopic heterogeneity.

Acupoints are modelled as regions of unusually high **metric anisotropy**:

$$\mathcal{A}_{\text{aniso}}(z) = \frac{\lambda_{\max}(g(z))}{\lambda_{\min}(g(z))}. \quad (8)$$

OGM implication: Optogenetic stimulation can verify the fixedness and robustness of meridian pathways; acupoints are geometric sensitive points for light stimulation.

2.3.4 2.4 Pathological Layer: Disease as Topological Collapse

Health is characterised by **multiple coexisting attractors**: the system cycles among functional states within the healthy basin $\mathcal{B}_{\text{healthy}}$. Pathology corresponds to **topological collapse**: the attractor structure is compressed into a single pathological attractor, and the system becomes trapped in $\mathcal{B}_{\text{pathological}}$.

The separatrix is the geometric threshold: when external perturbation exceeds the barrier height, the state vector crosses from the healthy domain into the pathological domain.

OGM implication: Optogenetic stimulation can reverse pathological topological collapse by restoring the attractor landscape—increasing the depth of healthy basins and lowering pathological barriers.

2.3.5 2.5 Memory Layer: Memory as Curvature-Stabilised Attractors

Immune memory and neural memory share a common geometric structure: **curvature-stabilised attractors**. Memory stability is determined by basin depth: negative curvature stabilises memory, positive curvature may erase it.

The geometric definition of memory: an attractor basin on the state manifold, defined by curvature, capable of integrating discrete inputs. The topological protection that hides the missing 70% of the proton' s spin (a global SU(3) topological invariant) is fundamentally linked to the stability of memory attractors.

OGM implication: Optogenetic stimulation can write, stabilise, or erase memory by changing local curvature.

2.4 3. First-Principles Derivation of the Structural Isomorphism

We begin by stating precisely what can, and cannot, be derived. The claim that cosmic curvature and physiological curvature are *numerically identical* ($R_{\text{cosmic}} = R_{\text{physio}}$) is not derivable from first principles and is not asserted here; it would require a bridge law whose independent justification is presently lacking. What *can* be derived is a **structural isomorphism**: both quantities are the same functorial construction applied to two different objects—namely, the curvature invariant ring of a Hessian geometry generated by an extremal-information principle. The derivation below makes this claim exact and enumerates the conditions under which it fails.

2.4.1 3.1 Axioms

The derivation rests on three axioms chosen because they are shared by both domains and make no reference to domain-specific ontology (no metric tensor, no action potential, no Lagrangian):

(A1) States are distributions. Any physically statable macroscopic condition of a system is equivalent to a family of probability distributions $\{p(x \mid \theta)\}$ over micro-configurations x, parametrised by a set of distinguishable control parameters or macroscopic observables $\theta = (\theta^1, \dots, \theta^n)$. The state space is the parameter manifold $\mathcal{M}$.

(A2) Maximum entropy. Subject to constraints $\{\langle f_a \rangle = c_a\}$, the realised distribution maximises Shannon entropy $S = -\int p \ln p$, yielding the Gibbs form $p = Z^{-1} \exp(-\sum_a \lambda_a f_a)$ with $Z(\lambda)$ the partition function and $c_a = -\partial_{\lambda_a} \ln Z$.

(A3) Observational composability. Coarse-graining and reparametrisation do not alter the informational content of an inference (sufficiency principle): if $T(x)$ is sufficient for θ, then all inferential statements factor through T.

Axioms (A1)-(A3) hold for the vacuum state of a quantum field near a causal horizon and for a multi-scale physiological ensemble alike; neither general covariance nor cellular biology enters.

2.4.2 3.2 The Metric Is Forced, Not Chosen

From (A2), the entropy and the log-partition function are Legendre dual, and

$$g_{ab}(\theta) \;\equiv\; \frac{\partial^2 \ln Z}{\partial\lambda^a \partial\lambda^b} = \langle f_a f_b \rangle - \langle f_a \rangle \langle f_b \rangle = -\frac{\partial^2 S}{\partial\theta^a \partial\theta^b}\Big|_{\text{eq}}.$$

Thus $g = \operatorname{Hess} S$. Crucially, this identification is not a modelling convenience. By the Chentsov theorem—the unique characterisation theorem of information geometry—any Riemannian metric on $\mathcal{M}$ invariant under sufficient statistics (Markov embeddings) is, up to a constant factor, the Fisher metric g. Hence:

Theorem 1. *For any system satisfying (A1)-(A3), the state manifold $(\mathcal{M}, g)$ is uniquely determined and $g = \operatorname{Hess} S$.*

Consequently, the existence of a metric—and therefore of a Levi-Civita connection, a Riemann tensor, and a scalar curvature R—is a consequence of the axioms alone. At this stage the cosmic and physiological cases are not merely analogous; they are instances of the same construction.

2.4.3 3.3 Curvature as Inferential Path Dependence

The physical meaning of R follows from its operational definition. Parallel transport of an estimator around a closed loop in $\mathcal{M}$ produces a net displacement proportional to the curvature tensor; equivalently, R is the irreducible part of the fourth-order term in the expansion of the Kullback-Leibler divergence $D_{\mathrm{KL}}(p_\theta \| p_{\theta+d\theta}) = \frac{1}{2} g_{ab} d\theta^a d\theta^b + O(|d\theta|^4)$ that cannot be removed by any reparametrisation. Therefore:

Proposition 2. *R is the intrinsic invariant of local distinguishability of neighbouring states. $R = 0$ iff local measurements can be patched globally without contradiction (integrable, memoryless, Markovian); $R \neq 0$ iff inference is path-dependent, i.e. the system carries intrinsic geometric memory.*

This renders the intuition "memory is curvature" a theorem rather than a metaphor, and it explains why curvature is the natural carrier of cross-scale information: it is invariant under the choice of coordinates and of measurement apparatus, depending only on the distribution family itself.

2.4.4 3.4 Scale Invariance from Fixed Points, Not from Coincidence

The original scale-invariance hypothesis is replaced here by a derivable statement. Let $\mathcal{R}_s : \mathcal{M} \to \mathcal{M}$ be the coarse-graining operator at scale s, and let θ^* be a fixed point, $\mathcal{R}_s\theta^* = \theta^*$. Linearising in its neighbourhood gives eigen-directions with scaling exponents y_a. Then:

Theorem 3. *At a critical fixed point the correlation length diverges, $\xi \to \infty$, the system loses all intrinsic scales, and the only remaining length is the curvature radius $\ell_R \sim |R|^{-1/2}$. The metric can therefore transform only by an overall conformal factor, $g \mapsto s^{2\Delta} g$, hence $R \mapsto s^{-2\Delta} R$.*

Scale invariance is thus a *fixed-point property*, not a numerical identity between two curvatures. On the physiological side, living systems ubiquitously exhibit signatures of self-organised or tuned criticality—neuronal avalanches, heart-rate variability, gene-expression noise—indicating that the physiological manifold likewise lies near a critical fixed point. Power laws appear in both domains because they are two instances of the same mechanism, without requiring mutual causation.

2.4.5 3.5 The Spacetime Side: Curvature as a Response Function of the Vacuum State Manifold

To close the argument on the cosmic side we invoke established results linking entropy variation to gravitational dynamics. Jacobson showed that imposing the Clausius relation $\delta Q = T\delta S$ on all local Rindler horizons yields the Einstein field equations. The entropy in question is the vacuum entanglement entropy, and its second-order response to deformations is a quadratic form—precisely the Fisher metric on the family of vacuum states. Hence:

Theorem 4. *Spacetime curvature is the response function of the information metric on the vacuum-state manifold, and the Einstein equation is the extremality condition of that metric.*

Combined with Theorem 1, this establishes that R_{cosmic} is itself an invariant of $\operatorname{Hess} S$; it is not a geometric object independent of information geometry.

2.4.6 3.6 The Physiological Side: The Nonequilibrium Generalisation

The physiological case differs in three essential respects that must be stated explicitly: its symmetry algebra is low and broken rather than highly symmetric; it is sustained far from equilibrium; and its arrow of time is maintained actively by dissipation. Accordingly the equilibrium entropy S must be replaced by a nonequilibrium potential Φ (the Freidlin–Wentzell quasipotential, or large-deviation rate function), and the metric becomes

$$g_{ab} = \frac{\partial^2 \Phi}{\partial\theta^a \partial\theta^b},$$

with curvature still Hessian-generated but no longer simplified by symmetry.

Proposition 5. *R_{physio} is the intrinsic invariant of the nonequilibrium potential landscape, quantifying the path dependence of transitions between steady states. A merger of attractor basins corresponds to a change in the Morse index of Φ—a topological event—at which curvature necessarily diverges or jumps.*

This is the rigorous form of the "disease as topological collapse" postulate: collapse is a bifurcation in Φ, detected geometrically by singular behaviour of R.

Geometric meaning of optogenetic control. Light stimulation alters the open probability of ion channels, thereby changing the Hessian structure of the effective potential Φ in a local region of the state manifold. Geometrically, this is equivalent to injecting a controllable curvature perturbation δR into the manifold. The essence of optogenetic control is therefore: **precise programming of the local curvature field $R_{\mathbf{physio}}(z,t)$ via light pulses.**

2.4.7 3.7 Statement of the Isomorphism

Let $\mathfrak{S}$ be the category whose objects are triples $(\mathcal{M}, g, \mathcal{C})$ with $g = \mathrm{Hess}\,\Phi$ generated by an extremal-information principle and $\mathcal{C}$ the constraint algebra (symmetry group and its representation class), and whose morphisms are smooth maps preserving the extremal structure of Φ and the representation class of $\mathcal{C}$. Define the curvature functor $\mathcal{R} : \mathfrak{S} \to \mathbf{Inv}$ sending an object to the ring generated by scalar curvature spectrum, Ricci eigenvalues, Chern classes, and Euler characteristic. Let $\mathcal{O}_c = (\mathcal{M}_c, g_c, \mathcal{C}_c)$ and $\mathcal{O}_p = (\mathcal{M}_p, g_p, \mathcal{C}_p)$ denote the cosmological and physiological objects, both generated by (A1)-(A3).

Isomorphism Theorem (structural form). *There exists a canonical isomorphism of invariant rings*

$$\Psi : \ \mathcal{R}(\mathcal{O}_c)/\!\sim \ \xrightarrow{\cong} \ \mathcal{R}(\mathcal{O}_p)/\!\sim,$$

i.e. at the level of the ring generated by scalar curvature, Ricci eigenvalues, Chern classes, and Euler characteristic—all derived from $\mathrm{Hess}\,\Phi$*—the cosmic and physiological cases are isomorphic. The isomorphism arises because both are produced by the same construction functor, not because of empirical fitting.*

Non-isometry clause (essential). Ψ *does not in general lift to an isometry* $(\mathcal{M}_c, g_c) \cong (\mathcal{M}_p, g_p)$*:* $\mathcal{C}_c \neq \mathcal{C}_p$ *(different symmetry algebras),* $\dim \mathcal{M}_c \neq \dim \mathcal{M}_p$*, one side is equilibrating and the other dissipative, and no bridge law fixes a common numerical scale. Consequently the strongest defensible claim is: the two systems share the same grammar but write different sentences.*

2.4.8 3.8 Corollaries –Testable Consequences Independent of the Original Hypothesis

The structural isomorphism yields four predictions that do not depend on assuming $R_{\mathrm{cosmic}} = R_{\mathrm{physio}}$:

(i) Curvature–fluctuation–response triad. For any $\mathrm{Hess}\,\Phi$ system, $\chi_{ab} = \partial\langle O_a\rangle/\partial h_b = \beta g_{ab}$, hence R is reconstructible from macroscopic response functions alone (compliance, gain, sensitivity) without microscopic access. *Test:* reconstruct g and R from heart-rate variability, functional connectivity, or lung compliance data and regress against outcomes; failure of R to outperform raw variables falsifies the architecture.

(ii) Unified critical scaling. Near a fixed point, $C(r) \sim r^{-(d-2+\eta)}$, $\xi \sim |\theta - \theta_c|^{-\nu}$, and necessarily $R \sim \xi^{-2} \sim |\theta - \theta_c|^{2\nu}$. *Test:* measure ν independently at tissue scale and check whether $R\xi^2$ is a universal constant.

(iii) Topological protection criterion. Protected edge channels require integer bulk invariants and an open gap. *Test:* identify transport modes in neural or cardiac tissue that are insensitive to microscopic heterogeneity (drug dose, cell-to-cell variability) but switch abruptly at global topological parameters (connectivity number, winding number); absence of gap-protected plateaux undermines the topological interpretation of meridians.

(iv) Geometric law of memory decay. Geodesic deviation obeys $\ddot{J} + RJ = 0$, so recovery dynamics within an attractor basin satisfy $\tau \sim |R|^{-1/2}$, with $R < 0$ implying exponential sensitivity and forgetting, and $R > 0$ implying oscillation and retention. *Test:* fit τ from sleep-dependent memory consolidation or LTP experiments against an independently estimated R. This is the cleanest and most readily falsifiable item in the entire chain.

2.4.9 3.9 Conditions Under Which the Derivation Collapses

A architecture' s value lies in how it can be overturned. The present derivation fails if any of the following holds:

1. **The sufficiency structure of physiological observables invalidates the Fisher metric** (e.g. the observation operator itself depends on θ), so Theorem 1 loses uniqueness and the construction becomes a convention.
2. **The entropy–gravity bridge fails** (the locality/equilibrium or entanglement-extremality premises do not survive in the correct quantum-gravitational description), so Theorem 4 fails and R_{cosmic} reverts to being an entity independent of information geometry, severing the isomorphism.
3. **Healthy physiological states are empirically deep in a non-critical regime** (exponential cut-offs, well-defined characteristic scales), so Theorem 3' s premise fails and scale invariance returns to being an unproven assumption.

All three are empirically decidable.

2.5 4. Core Architecture: Optogenetic Geometric Medicine

2.5.1 4.1 Core Proposition

Optogenetic control is the injection of a controllable perturbation δf^{μ} into the physiological state manifold. By changing local curvature R, this perturbation reprograms the attractor landscape, guiding the system between healthy domains, pathological domains, and topologically protected edge states.

2.5.2 4.2 The Five-Step Geometric Engineering Protocol

Step	Operation	Mathematical Formulation	Corresponding Technology
1. Map the manifold	Reconstruct the physiological state manifold	Estimate $(\mathcal{M}, g)$ from multimodal data	Optogenetics + multi-electrode recording + calcium imaging
2. Measure geometry	Estimate local curvature and topology	Compute $R(z)$, $\mathcal{A}_{\text{aniso}}$, Chern number	Manifold learning + topological data analysis
3. Program perturbation	Design light pulse patterns	Construct $\delta f^{\mu}(t)$	Optogenetic stimulation protocols
4. Reprogram landscape	Change basin depths and barriers	Adjust $\Phi(z)$ and $R(z)$	Closed-loop optogenetic control

Step	Operation	Mathematical Formulation	Corresponding Technology
5. Verify topology	Test whether invariants change	Compute $\mathcal{C}$, χ, H_1	Persistent homology + topological invariant estimation

2.5.3 4.3 Causal Chain: From Light Pulse to Geometric Flow

The causal chain from light stimulation to physiological response:

Light pulse → Channelrhodopsin activation → Membrane potential change → Local charge redistribution → Loca

Key equations:

- Light stimulation rewrites drift term: $f^\mu \to f^\mu + \delta f^\mu(z, t; x_a)$
- Curvature change alters decoherence: $\gamma(R) \propto R^{3/2}$
- Attractor basin depth change: $\Delta\Phi \propto \Delta R$

2.5.4 4.4 Connection to the Structural Isomorphism

Section 3 established the structural isomorphism between cosmic and physiological curvature. The key implication for OGM is: Every optogenetic stimulation experiment is a physiological-scale test of the same geometric grammar that governs cosmological curvature dynamics. If OGM observes curvature-dependent thresholds, nonlinear responses, and topological protection at the physiological scale, then the structural isomorphism' s predictions are verified at a scale accessible to laboratory experiment. Conversely, if OGM fails to observe these signatures, the physiological side of the isomorphism is falsified.

The isomorphism does not require $R_{\text{cosmic}} = R_{\text{physio}}$. It requires only that both curvatures are generated by the same Hessian construction, and therefore share the same invariant ring structure. This is what makes the architecture testable: we test the grammar, not the numerical value.

2.6 5. Step-5 Protocol: Optogenetic Geometric Reprogramming of Physiological State Manifolds

This section advances the curvature scale-invariance hypothesis and the structural isomorphism theorem established in the preceding sections from a descriptive geometric architecture into an intervenable, optimizable, and falsifiable dynamical protocol. The central claim is that physiological systems—neural population states, cell-fate landscapes, and disease configurations—can all be understood as attractor structures embedded on state manifolds. Therapeutic intervention or cellular reprogramming should therefore not be conceived as merely pushing a variable back to a nominal range, but as reshaping the geometry and topology of the attractor landscape so that the system spontaneously stabilises in a healthy attractor. Optogenetics, deep brain stimulation, transcranial magnetic stimulation, transcriptional reprogramming, and closed-loop neuromodulation can all be unified as controllable perturbations δf^μ acting on the underlying dynamical vector field.

2.6.1 5.1 Physiological State Manifolds and Geometric Metric

Let the physiological state space be a differentiable manifold $\mathcal{M}$ with local coordinates z^μ, which may include membrane potentials, ion concentrations, gene-expression levels, metabolites, heart-rate variability features, BOLD signals, or low-dimensional latent variables extracted from neural population activity. In the absence of external intervention, the system evolves according to

$$\dot{z}^\mu = f^\mu(z) + \xi^\mu(t), \quad (9)$$

where $\xi^\mu(t)$ represents intrinsic noise and environmental fluctuations. If the system admits a non-equilibrium potential or quasi-potential $\Phi(z)$, the deterministic component can be decomposed as

$$f^\mu(z) = -g^{\mu\nu}\partial_\nu\Phi(z) + J^{\mu\nu}\partial_\nu\Phi(z), \quad (10)$$

where the first term is a gradient flow driving the system toward regions of lower potential, and the second term is a non-conservative flow sustaining oscillations, cycles, and far-from-equilibrium steady states. Healthy states correspond to deep attractor basins of Φ, whereas disease states correspond to abnormally deep, abnormally shallow, abnormally isolated, or topologically misplaced attractor basins.

The metric tensor $g_{\mu\nu}$ need not be prescribed a priori. It can be obtained through three complementary constructions:

1. **Information geometry.** The metric is defined by the Fisher information matrix of the state distribution,

$$g_{ab}(z) = \mathbb{E}\left[\partial_a \log p(x|z)\, \partial_b \log p(x|z)\right]. \quad (11)$$

This is the unique metric (up to a constant factor) invariant under sufficient statistics, as guaranteed by Chentsov' s theorem.

2. **Response geometry.** The metric is estimated from linear response functions or generalised susceptibilities, quantifying how sensitive each state variable is to small external perturbations.

3. **Dynamical geometry.** The metric is constructed from the local Jacobian and noise covariance,

$$g_{\mu\nu} \sim \left(\mathrm{Hess}\,\Phi + \text{antisymmetric correction}\right)_{\mu\nu}. \quad (12)$$

Under this construction, curvature $R[g]$ becomes a geometric proxy for physiological plasticity rather than a purely mathematical ornament. Regions of positive curvature tend to preserve neighbouring trajectories and stabilise memory-like patterns; regions of negative curvature amplify small perturbations and facilitate state transitions; near-zero or scale-invariant curvature may correspond to critical regimes, scale-free coordination, or cross-scale integrative states.

2.6.2 5.2 Optogenetic Perturbation as Vector-Field Editing

The distinctive value of optogenetics lies in its ability to convert "stimulation" from an ambiguous external input into a structured edit of the dynamical vector field. Suppose the system contains K targetable cell types or regulatory modules. The intervened dynamics are then written as

$$\dot{z}^{\mu} = f^{\mu}(z) + \sum_{k=1}^{K} u_k(t)\, X_k^{\mu}(z) + \xi^{\mu}(t). \quad (13)$$

Here $u_k(t)$ denotes the stimulus intensity or control signal, and $X_k^{\mu}(z)$ is the perturbation vector field associated with the k-th target. The structure of X_k^{μ} is determined by the expression pattern of the optogenetic actuator, cell-type identity, synaptic connectivity, ion selectivity, and downstream network architecture. Blue-light activation of ChR2-like cation channels produces depolarising current injection; yellow-light activation of inhibitory rhodopsins produces hyperpolarisation or synaptic suppression; multi-colour, multi-target stimulation generates linear or nonlinear combinations of perturbation fields.

Within the broader geometric medicine architecture, optogenetics is one high-temporal-precision instantiation of δf^{μ}. Deep brain stimulation, transcranial magnetic stimulation, vagus nerve stimulation, pharmacological modulation, transcriptional reprogramming, and metabolic interventions can all be expressed in the same form, differing primarily in spatial resolution, temporal precision, reversibility, and biological constraints.

2.6.3 5.3 Reprogramming the Attractor Landscape

Let $\mathcal{A}_H$ denote the set of healthy attractors and $\mathcal{A}_D$ the set of pathological attractors. The reprogramming task is not to instantaneously displace the system into $\mathcal{A}_H$, but to reshape the landscape so that $\mathcal{A}_D$ is shallowed, fragmented, or eliminated, while $\mathcal{A}_H$ acquires sufficient basin depth and appropriate accessibility. Formally, the post-intervention potential Φ' should satisfy

$$\Phi'(z) \gg \Phi(z) \quad \text{for } z \in \mathcal{N}(\mathcal{A}_D), \quad (14)$$

while preserving necessary physiological oscillations, adaptive responses, and non-conservative flow structures.

Curvature plays three roles in this reprogramming:

- **Stability indicator.** Curvature changes near attractor basin boundaries reflect the difficulty of state switching.
- **Plasticity indicator.** Negative-curvature channels correspond to low-cost migration paths between states.
- **Memory-retention indicator.** Positive-curvature regions correspond to stable patterns resistant to noise or brief perturbations.

Therapeutic intervention can thus be geometrically formulated as redirecting probability flow from $\mathcal{A}_D$ to $\mathcal{A}_H$ along negative-curvature channels, without destabilising the system' s global robustness.

2.6.4 5.4 Optimal Reprogramming Trajectories

Under Freidlin–Wentzell large-deviation theory, rare noise-driven transitions from a pathological attractor to a healthy attractor are governed by the action functional

$$S[z] = \frac{1}{2}\int_0^T \left\| \dot{z}^\mu - f^\mu(z) - \sum_k u_k(t) X_k^\mu(z) \right\|_{D^{-1}}^2 dt. \quad (15)$$

The optimal intervention problem can be formulated as

$$\min_{u(t),\, z(t)} \int_0^T \left[\alpha\, \mathcal{R}(z, g, R) + \beta\, \mathcal{C}_{\text{stab}}(z) + \gamma\, \|u(t)\|^2\right] dt, \quad (16)$$

subject to

$$z(0) \in \mathcal{B}(\mathcal{A}_D), \qquad z(T) \in \mathcal{B}(\mathcal{A}_H), \quad (17)$$

$$\dot{z}^\mu = f^\mu(z) + \sum_k u_k(t) X_k^\mu(z), \quad (18)$$

and physiological safety constraints:

$$0 \le u_k(t) \le u_k^{\max}, \quad \dot{z}^\mu \in \mathcal{S}_{\text{safe}}, \quad \Phi(z(T)) \ge \Phi_{\min}. \quad (19)$$

Here $\mathcal{R}(z, g, R)$ may be a curvature regulariser constructed from Ricci scalar curvature, sectional curvature, or Ollivier–Ricci curvature on functional or physiological networks. $\mathcal{C}_{\text{stab}}$ may encode attractor basin depth, Lyapunov exponents, or transition-rate constraints. This optimisation naturally falls within the algorithmic architecture for optimal trajectory solving in complex systems: the state space carries a nontrivial metric, the dynamics include both conservative and non-conservative components, noise is irreducible, and control inputs are constrained by biophysical limits.

2.6.5 5.5 Observable Geometric Quantities and Experimental Readouts

To avoid remaining purely formal, the protocol requires estimable readouts for curvature, attractors, and vector-field perturbations.

Data modality	Estimable geometric quantity	Intervention validation
Calcium imaging / electrophysiology	Neural population manifold, local curvature, geodesic deviation	Pre/post optogenetic manifold geometry
fMRI / EEG	Functional-connectivity curvature, Ollivier–Ricci curvature, transition probabilities	Network robustness under closed-loop stimulation

Data modality	Estimable geometric quantity	Intervention validation
Single-cell transcriptomics	Cell-fate manifold, Waddington potential, branch-point curvature	Reprogramming factor or optogenetic pathway perturbation
Heart-rate variability / physiological signals	Physiological state distribution, Fisher information, response manifold	Stability change after autonomic or brain stimulation
Behavioural readouts	Attractor dwell time, state-switching rate	Whether behaviour enters and remains in a healthy attractor

The key falsifiability criteria are not whether stimulation changes a single variable, but whether:

1. **Stimulation alters local manifold curvature in the predicted direction;**
2. **Attractor basin depths or boundary positions shift as predicted;**
3. **State migration follows the predicted negative-curvature channel;**
4. **The system remains in the healthy attractor after stimulation is withdrawn;**
5. **The intervention cost approaches the lower bound predicted by the optimal trajectory.**

2.6.6 5.6 Closure with the Structural Isomorphism Theorem

Section 3 established a structural isomorphism between the cosmic curvature scale R_{cosmic} and the physiological curvature scale R_{physio}. This section completes that isomorphism dynamically. If R_{cosmic} characterises a fixed-point relation between observational scale and curvature radius in spacetime geometry, then R_{physio} characterises a corresponding fixed-point relation among perception, regulation, and stability in physiological systems. Optogenetic perturbation δf^{μ} performs a local geometric edit on the physiological manifold, steering the system back toward a coordinated curvature–scale–stability regime.

Accordingly, the Step-5 protocol is not an appended application chapter but the dynamical completion of the isomorphism theorem:

> R_{cosmic} specifies the geometric structure of scale invariance; R_{physio} realises that structure within physiological systems; optogenetic geometric medicine provides the operational rules for controllable reprogramming on R_{physio}.

2.7 6. Experimental Validation

2.7.1 6.1 Curvature-Dependent Threshold Behavior

OGM prediction: Dynamical transitions on the physiological state manifold exhibit thresholds determined by local curvature.

Experimental evidence:

Teplenin et al. demonstrated that in optogenetically modified monolayers of cardiac cells, ectopic excitation originates at areas of **maximal curvature** of the depolarised region boundary—where stimulating electrotonic currents are **minimal**. This paradoxical finding—that excitation occurs

where stimulus is weakest—cannot be explained by standard critical nucleation theory. It requires curvature-dependent instability at the boundary.

More quantitatively, Connolly et al. showed that ventricular endocardial ridges (surfaces of **negative mean curvature**) had a stimulus capture threshold **0.21 ± 0.03 V lower** than endocardial grooves (surfaces of **positive mean curvature**)—a **24% reduction** in threshold, corresponding to **56.2 ± 6.4% of the energy**. This provides direct quantitative evidence that **curvature determines threshold**.

Parameter	Ridge (negative curvature)	Groove (positive curvature)	Difference
Capture threshold	Lower	Higher	0.21 ± 0.03 V
Energy required	56.2 ± 6.4%	100%	~44% reduction
ERP at minimal stimulus	Shorter	Longer	7.4 ± 4.2 ms
ERP at identical stimulus	Shorter	Longer	15.8 ± 5.3 ms

Conclusion: Curvature determines threshold—the OGM prediction is directly verified at the physiological scale.

2.7.2 6.2 Nonlinear Response

OGM prediction: Optogenetic response function follows $\Phi_{AB}(I) \propto (I - I_{\text{th}})^{\beta}$, with nonlinear exponent $\beta > 1$.

Experimental evidence:

Jia et al. demonstrated that optogenetic activation of inhibitory interneurons **cannot suppress but enhances** the frequency and synchronisation of spiking of excitatory pyramidal neurons—a phenomenon called synchronised post-inhibitory rebound (PIR) spiking. This paradoxical nonlinearity arises from complex interactions between hyperpolarisation-activated cation currents and membrane potential. Strong optogenetic stimulation with high spiking frequency contributes to PIR spiking.

In cortical cultures, frequency-optimised optogenetic stimulation revealed that network-level potentiation depends non-monotonically on stimulation frequency, with optimal frequencies rather than monotonic increases.

Temporal precision studies showed that ChroME2s-expressing neurons maintain single-spike resolution up to **80 Hz**, while CoChR-expressing neurons maintain up to **60 Hz**, with multiple spikes emerging beyond 60 ms pulse width due to cumulative depolarisation and plateau potentials.

Conclusion: Nonlinear response is a universal feature of optogenetic control—the OGM prediction is phenomenologically confirmed.

2.7.3 6.3 Topological Protection

OGM prediction: Topologically protected edge states exhibit extreme robustness against local perturbations.

Experimental evidence:

Ori et al. demonstrated that gap junction-coupled interfaces between **non-excitable tissues can be electrically excitable**—a phenomenon they termed **topologically stabilised action potentials**. This topological excitability occurs over a **far larger range of ion channel expression levels** than does excitability in the bulk.

The mechanism is geometric: across a gap junction-coupled interface between tissues with different resting potentials, membrane potential must pass through all intermediate values—including regions where it would be unstable in the bulk. This continuity requirement leads to topologically protected excitations at tissue interfaces.

Furthermore, the non-monotonic dependence of optogenetic control efficiency on illumination area persists across molecular, cellular, microstructural, and tissue scales. This multi-scale invariance underscores that the optimal illumination range is **not a model-specific artifact, but an emergent property rooted in the universal physics of nonlinear wave dynamics**.

Conclusion: Topological protection mechanism exists in real physiological systems—the OGM prediction has a biological precedent.

2.7.4 6.4 Summary of Experimental Validation

OGM Prediction	Experimental Evidence	Reference
Curvature determines threshold	Ridge threshold 24% lower than groove; ectopic excitation at maximal curvature	Teplenin et al.; Connolly et al.
Nonlinear response	Non-monotonic frequency dependence; PIR spiking; frequency-optimised potentiation	Jia et al.; cortical cultures
Topological protection	Topological action potentials at non-excitable interfaces; multi-scale invariance	Ori et al.
Curvature-dependent decoherence	$T_2 \propto R^{-3/2}$ (predicted, not yet directly measured)	To be tested

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2.8 7. Testable Predictions

2.8.1 7.1 Quantitative Predictions

Prediction	Mathematical Formulation	Experimental Protocol
Curvature-dependent threshold	$I_{\text{th}} \propto \Delta R$	Measure optogenetic threshold across regions of varying curvature
Nonlinear response	$\Phi_{AB}(I) \propto (I - I_{\text{th}})^{\beta}$	Measure intensity-response curves; extract β

Prediction	Mathematical Formulation	Experimental Protocol
Decoherence scaling	$T_2 \propto R^{-3/2}$	Measure coherence time in regions of varying curvature
Frequency shift	$\Delta\omega \propto R^{1/4}$	Measure neural oscillation frequency shift under optogenetic stimulation
Topological robustness	$\mathcal{C}$ invariant under local perturbation	Measure topological invariants before and after localised injury
Memory decay law	$\tau \sim \lvert R\rvert^{-1/2}$	Fit recovery time from LTP experiments against independently estimated R
Curvature-fluctuation-response	$\chi_{ab} = \beta g_{ab}$	Reconstruct g and R from heart-rate variability; regress against outcomes

2.8.2 7.2 Critical Experiment: Non-Local Coupling

Prediction: Anatomically distant but geometrically close ($d_g \ll 1$) point pairs should exhibit non-local coupling determined by state-space geodesic distance, not anatomical distance.

Protocol: Optogenetically stimulate an acupoint on the foot; measure cardiac response. If response strength is determined by state-space geodesic distance d_g rather than anatomical distance, the OGM prediction is decisively verified.

2.8.3 7.3 Phase Locking

Prediction: Optimal light stimulation windows are determined by the phase relationship between the system' s intrinsic period and curvature fluctuations.

Protocol: Apply optogenetic stimulation at different circadian phases; measure phase-dependent response strength.

2.8.4 7.4 Curvature Scale Invariance Test

Prediction: The same mathematical structures (curvature-dependent thresholds, nonlinear responses, topological protection) observed at physiological scales should mirror those predicted at cosmological scales.

Protocol: Compare optogenetic response statistics with MES cosmology predictions for curvature-driven dynamics; test whether the same scaling exponents $\gamma \propto R^{3/2}$, $\Delta\omega \propto R^{1/4}$ appear in both domains.

2.9 8. Discussion

2.9.1 8.1 Relationship to Existing Architectures

OGM does not replace existing optogenetics research but provides a unified geometric language. It complements:

- **State-space optimal feedback control**: Adds geometric-topological interpretation layer
- **Neural manifold theory**: Connects manifold structure to optogenetic control
- **Optogenetic application studies**: Provides theoretical architecture for interpreting accumulated data

2.9.2 8.2 What the Structural Isomorphism Does and Does Not Claim

The derivation of Section 3 makes precise what can be asserted:

- **It does claim**: a canonical isomorphism of curvature-invariant rings generated by a Hessian geometry.
- **It does not claim**: numerical identity $R_{\text{cosmic}} = R_{\text{physio}}$, or an isometry between the two manifolds.

This distinction is essential for the architecture' s scientific integrity. The isomorphism is structural, not numerical; it is derived, not assumed; it is falsifiable, not tautological.

2.9.3 8.3 Limitations and Challenges

Challenge	Description	Mitigation Strategy
Mathematical barrier	Riemannian geometry, topological invariants, SDEs	Present geometric intuition + testable predictions in main text; rigorous derivations in Section 3
Curvature estimation	High-dimensional manifold curvature estimation is noise-sensitive	Uncertainty quantification + bootstrap analysis
Scale distinction	Cosmological and physiological curvatures are different scales	Strictly distinguish symbols; emphasise structural isomorphism, not numerical identity
Topological invariant computation	Computing Chern numbers from finite data is difficult	Validate on synthetic data + persistent homology
Sufficiency structure	If observation operator depends on θ, Theorem 1 loses uniqueness	Empirically test whether physiological observables satisfy (A1)-(A3)

2.9.4 8.4 Future Directions

1. **Geometric optogenetics**: Use light pulses to precisely program geodesic trajectories on the manifold
2. **Topological optogenetics**: Use light stimulation to probe and control topologically protected edge states

3. **Quantum optogenetics**: Verify curvature-driven Lindblad dynamics at the quantum scale
4. **Isomorphism verification**: Test the four corollaries of Section 3.8 independently of the full isomorphism hypothesis

2.9.5 8.5 Ultimate Vision

Optogenetic Geometric Medicine elevates optogenetics from a "neural control tool" to a "geometric engineering platform for physiological state manifolds."It provides a programmable, observable, and repeatable experimental basis for the paradigm shift from substance-based medicine to relationship-based medicine. In the laboratory, light pulses replay the geometric breathing of the cosmos—testing the structural isomorphism hypothesis at the physiological scale.

2.10 9. Conclusion

We have proposed Optogenetic Geometric Medicine (OGM), a unified architecture that reinterprets optogenetics as curvature-driven reprogramming of physiological state manifolds. The core findings are:

1. **Structural isomorphism derived from first principles**: Cosmological and physiological curvatures are the same functorial construction—the curvature-invariant ring of a Hessian geometry generated by an extremal-information principle—applied to two different objects. This is a theorem, not a hypothesis.

2. **Light stimulation rewrites drift, not merely ion flux**: Optogenetic control injects a perturbation δf^{μ} that changes local curvature R, thereby reprogramming the attractor landscape.

3. **Shared geometric architecture**: Neural memory, immune memory, pathological states, and meridian networks share the same geometric structures—attractors, separatrices, and topologically protected edge states.

4. **Step-5 Optogenetic Geometric Reprogramming Protocol**: The architecture is completed dynamically by a constrained optimisation protocol that uses Freidlin–Wentzell large-deviation theory to compute optimal reprogramming trajectories, with five explicit falsifiability criteria and a comprehensive table of observable geometric quantities.

5. **Experimental validation**: Curvature-dependent thresholds, nonlinear responses, and topological protection have been verified in optogenetic cardiac and neural systems.

6. **Four testable corollaries**: The curvature–fluctuation–response triad, unified critical scaling, topological protection criterion, and geometric law of memory decay are independent of the full isomorphism hypothesis.

7. **Three verification conditions**: The derivation collapses if the sufficiency structure of physiological observables invalidates the Fisher metric, if the entropy–gravity bridge fails, or if healthy physiological states are deep in a non-critical regime.

The OGM architecture provides a programmable, observable, and repeatable experimental platform for testing geometric medicine. It bridges quantum foundations (Lindblad dynamics), physiological organisation (meridian networks), pathological mechanisms (topological collapse), and cosmological principles (MES cosmology) into a unified architecture.

The ultimate significance of OGM depends not on whether traditional terminology can be given modern mathematical names, but on whether the resulting mathematical objects make successful, quantitative, and falsifiable predictions about living systems.

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2.13 Availability of data and material

Code, mathematical scripts, and preprints are available at GitHub: https://github.com/Zaitian001/-Self-Preprint-V1.1-

2.14 Competing interests

Not applicable

2.15 Authors Contributions

Baoliin (Zaitian) Wu 伍宝林 is the sole author of this manuscript and is responsible for all aspects of the research, including conceptualization, methodology, theoretical formulation, and manuscript writing.

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2.17 Physical Proof of Existence (Appendix)

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